

Nara Ávila

Master's Student

Department of Mathematical Physics
Physics Institute, University of São Paulo (USP)

E-mail: nara.moraes@usp.br

Website: <https://fma.if.usp.br/~avilamrs>

Education

M.Sc. in Nuclear Physics [\[link\]](#)

2025 – Present

Physics Institute (IF)

University of São Paulo (USP)

B.Sc. in Computer Engineering [\[link\]](#)

2006 – 2011

São Carlos School of Engineering (EESC)

University of São Paulo (USP)

The interdisciplinary curriculum encompassed four major areas of computer engineering:

Computer Science & Software Engineering: Training in algorithms and data structures, theory of computation, compilers, object-oriented programming, and distributed systems, complemented by mathematical programming and linear optimization. Software engineering covered requirements elicitation and specification, use-case analysis, UML models, operation contracts, the Unified Process, GRASP responsibility assignment, GoF design patterns, and MVC architecture. Software quality included inspections, structural testing based on control and data flow, and mutation testing, with supporting material on Java bytecode analysis using JaBUTi. Database studies addressed entity–relationship modeling, relational schema design, functional dependencies, normalization, relational algebra, and SQL. Performance evaluation combined experimental design, workload characterization, benchmarking, statistical analysis and confidence intervals with analytical and simulation models, including queueing networks and Petri nets. Multimedia coursework connected digital media representation with XML schemas and XSLT transformations for desktop and mobile presentation.

Computer Architecture: Study of digital logic, hardware description, microprocessors, and parallel computer organization, connecting circuit-level implementation with instruction execution and system performance. Topics included RISC and CISC instruction sets, addressing modes, pipelined execution, structural, data, and control hazards, forwarding, stalls, and branch handling. Superscalar and vector processing were examined alongside Flynn's taxonomy, SIMD and MIMD systems, and shared- and distributed-memory architectures. Hardware design used VHDL and programmable logic concepts. Digital integrated-circuit laboratory work included synthesis of a UART design with LeonardoSpectrum and AMS 0.35 μm standard-cell libraries, technology mapping, critical-path analysis, and area–speed optimization. The design flow extended to Verilog netlist export, schematic and layout generation in the Mentor environment, routing, and design-rule checking, linking behavioral descriptions to physical circuit implementation.

Physics & Electronics: Foundation in classical and quantum mechanics, electromagnetism, and experimental physics, integrated with circuit analysis, semiconductor devices, and analog and digital electronics. Device studies covered electronic states in crystals, intrinsic and doped semiconductors, p–n junctions and depletion regions, bipolar transistors and the Ebers–Moll model, MOS capacitors, and MOSFET current–voltage characteristics. Microelectronics connected CMOS inverter layout and process integration with silicon crystal growth, wafer preparation, photolithography, oxidation, wet and plasma etching, chemical vapor deposition, diffusion, ion implantation, and metallization. Signal processing addressed discrete-time signals and systems, sampling and quantization, the Z-transform, the discrete Fourier transform, and digital-filter design. Control studies connected dynamic-system models and feedback with digital implementation, analog-to-digital and digital-to-analog conversion, and microprocessor-based control of continuous physical processes.

Communications & Networks: Study of communication systems from electromagnetic propagation and signal representation to network protocols and multimedia delivery. Communication theory covered Fourier analysis, power spectral density, amplitude and angular modulation, noise, sampling, quantization, and bandwidth constraints. Antennas and optical communications connected radiation and transmission-line concepts with Maxwell's equations, wave propagation, and dielectric interfaces. Computer networking addressed layered architectures, data-link protocols, IP addressing and routing, ICMP, and TCP/UDP transport. Advanced topics included ATM service classes, SONET/SDH, cluster and storage interconnects, and quality of service through packet classification, scheduling, traffic policing, IntServ, and DiffServ. Practical material included Java video streaming with RTSP session control and RTP packetization over UDP, IEEE 802.11 frame analysis with Wireshark, and Android exercises involving Bluetooth discovery, Wi-Fi configuration, and network-traffic statistics.

Academic Positions

Research Collaborator [\[link\]](#)

2023 – 2025

40 hours/week

Project: Quantum Computing Applied to Molecular Modeling

Department of Materials Physics and Mechanics (DFMT)

Institute of Physics (IF), *University of São Paulo (USP)*

Research Context: Investigation of quantum computing methods for first-principles molecular modeling in materials and energy research. The work focused on computing ground-state energies of molecules relevant to the energy transition, including H₂ and CO₂, as a first step toward the use of quantum simulations in the discovery and development of new materials and sustainable energy technologies.

Quantum Simulation of Molecular Systems: Developed first-principles quantum simulations for molecular ground-state energy estimation. Electronic-structure problems were formulated within the Born–Oppenheimer approximation and second quantization, with fermionic Hamiltonians mapped onto qubit operators using the Jordan–Wigner transformation. Ground-state energies were estimated with VQE using UCCSD ansätze and benchmarked against Hartree–Fock and Full Configuration Interaction (FCI) calculations. The study examined basis-set selection, active-space approximations, shot counts, and qubit-reduction strategies, and explored restricted Boltzmann machines (RBMs) as alternative variational representations of molecular quantum states.

Quantum Computing Methods & Platforms: Implemented and evaluated gate-based quantum algorithms across Amazon Braket, IBM Qiskit, Google Cirq, CERN Qibo, and PennyLane, considering superconducting, trapped-ion, photonic, and neutral-atom architectures. Hardware-oriented studies examined execution time, noise, and decoherence, together with NISQ error mitigation using zero-noise extrapolation (ZNE) and Clifford data regression (CDR).

Teaching Assistant [\[link\]](#)

Fall 2009

8 hours/week

• SME0300 Numerical Methods I

Institute of Mathematics and Computer Sciences (ICMC)

University of São Paulo (USP)

Teaching Assistant [\[link\]](#)

Spring 2009

8 hours/week

• SME0601 Numerical Methods II

Institute of Mathematics and Computer Sciences (ICMC)

University of São Paulo (USP)

Academic Awards

• Best Presentation in Environmental Management System (SHS0436)

Fall 2009

- Recognized for an impactful final course presentation entitled “Harmonic and Disharmonic Systems,” due to its combination of visual simplicity and strong conceptual content. Using images alone, the presentation explored in detail the diversity, complexity, and beauty of harmonic systems in nature—including coral ecosystems and harmonious ecological relationships such as mutualism, commensalism, and epiphytism—in contrast with human systems that disrupt environmental balance, illustrated through examples of waste accumulation, pollution, environmental degradation, and pollution-related wildlife mortality, alongside other disharmonic relationships also found in nature, such as parasitism and disease. Received the maximum final grade (10.0/10.0).

- 9th place – ACM-ICPC / SBC Brazilian Programming Contest, First Phase [\[link\]](#) Sep 2007
 - Competed as a member of team “Greve de Acertos?” at the Campinas site of the first phase of the Brazilian Programming Contest (Maratona de Programação), the Brazilian regional competition of the ACM International Collegiate Programming Contest (ICPC).
- 6th place in the IPhO / SBF Brazilian Physics Olympiad [\[link\]](#) Oct 2003
 - National physics olympiad organized by the Brazilian Physical Society (SBF) and part of Brazil’s selection process for the International Physics Olympiad (IPhO).
- 18th place in the OBM / XI OMEG [\[link\]](#) Sep 2002
 - Mathematics olympiad competition associated with the Brazilian Mathematical Olympiad (OBM), Brazil’s national mathematics olympiad and part of the country’s selection pathway for the International Mathematical Olympiad (IMO).

Grants

AWS Cloud Credit for Research [\[link\]](#) Sep 2026
 Research Area: Quantum Computing
 Requested Amount: US\$5,000.00 in AWS Braket credits
 Status: Submitted September 17, 2026

EAP5054 Entrepreneurial Training for Scientists II (PRPG/USP) [\[link\]](#) Spring 2025

Project Title: StrategiQ — AI-Based SaaS for Quantum Computing Governance Consulting
 Role: Founder and Project Lead of a three-member team conducting market research and developing the business plan for StrategiQ developed through the Entrepreneurial Training for Scientists I (DPG5011)

Funding Agency: Office of the Provost for Graduate Studies, University of São Paulo (PRPG/USP)

Total Amount: R\$3,720.00

132734/2025-7 [\[link\]](#) 2025–2027

Program: Master’s Scholarship (GM)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Host Institution: University of São Paulo (USP)

Total Amount: R\$50,400.00

409642/2022-3 [\[link\]](#) 2023–2024

Project Title: Quantum Computing Applied to Molecular Modeling

Program: Technological Development in ICTs (DTC-F)

Funding Agency: National Council for Scientific and Technological Development (CNPq)

Scholarship Process: 301681/2023-6

Total Amount: R\$36,300.00

Software

1. MORAES, N. A.; LUZUM, M. W. *hic-bayes*. [\[Sec. 4.1\]](#) 2026
 Python software for Bayesian parameter estimation and Gaussian-process emulation in relativistic heavy-ioncollision studies. Implements PCA-based dimensionality reduction, Gaussian-process emulators, multivariate likelihoods, MCMC sampling, posterior analysis, and multi-fidelity workflows.
2. MORAES, N. A. *Zotero Mirroring Plugin*. [\[link\]](#) 2026
 Zotero plugin for mirroring the hierarchical structure of a Zotero library onto the local filesystem.
3. MORAES, N. A. *Git Release and Milestone CLI*. [\[link\]](#) 2026
 Command-line tool for automating Git releases, milestones, and software development cycles.
4. MORAES, N. A. *Quantum Battleship*. [\[link\]](#) 2025
 Quantum computing game that allows users to play Battleship using real quantum computers.
5. MORAES, N. A. *Molecular Symmetry*. [\[link\]](#) 2025

Web application for parsing molecular geometries from XYZ files and generating molecular symmetry reports.

Leadership Experience

Founder & Team Lead [\[link\]](#) *Fall 2025*

DPG5011 – Entrepreneurship, University of São Paulo (USP)

- Founded and led a six-member interdisciplinary team in the development of an innovative startup focused on quantum computing.
- Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.

Founder & Team Lead [\[link\]](#) *Spring 2025*

EAP5054 – Entrepreneurship, University of São Paulo (USP)

- Founded and led a three-member interdisciplinary team in the development of an innovative startup focused on quantum computing.
- Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.

Undergraduate Research Mentor *2025*

Institute of Physics, University of São Paulo (IFUSP)

- Mentored a group of five undergraduate students in scientific initiation projects in quantum computing, providing guidance throughout their research activities.
- Supervised the development of individual research projects presented by the students at the Undergraduate Research Symposium.

Applications & Proposals

Quantum Computing and Sustainability: Hackathon and Workshop [\[link\]](#) *Aug 2026*

- Application submitted for the Quantum Computing and Sustainability Hackathon at the Brazilian Center for Research in Physics (CBPF), Rio de Janeiro. Application under review.

Harvard Business School – MBA Program [\[link\]](#) *Jan 2023*

- Submitted application for admission to the MBA Program.

Professional Diplomas

Postgraduate Specialization in Financial and Capital Markets [\[link\]](#) *2021 – 2022*

- Quantitative methods, corporate finance, financing, and financial products and markets
- Mackenzie Presbyterian University*

Professional Experience

Software Engineering Specialist *2020 – 2022*

Open Co [\[link\]](#)

- Developed and optimized backend services for a digital banking platform, including the first installment payment solution via PIX.
- Proposed and prototyped the migration of a credit-analysis workload from Kubernetes (EKS) to AWS Lambda, reducing response time and operational costs by approximately 90%.

Senior Software Engineer *2019 – 2021*

TIDEXA [\[link\]](#)

- Contributed to the migration of a payroll system from Oracle Forms/PL–SQL to a Java-based platform.
- Led technical decisions and development activities within the engineering team.

Competitive Public-Sector Examination Preparation *2016 – 2018*

Independent, full-time study

- Multidisciplinary preparation for competitive Brazilian public-sector examinations, including Law, Accounting, Economics, Public Finance, Public Administration, and Business Administration.

Mid-Level Software Engineer	2014 – 2016
Itaotec – Grupo ITAUSA [link]	
Redesigned a business-critical software application using software design patterns and engineering best practices, improving usability, reliability, and maintainability. The application was deployed across more than 5,000 branches of a major Brazilian bank.	
Junior Software Engineer	2012 – 2014
Itaotec – Grupo ITAUSA	
Developed and maintained software applications for retail industry.	

Talks & Presentations

Conference Presentations

1. “The Effect of Correlated Experimental Uncertainties on Bayesian Inference of Quark–Gluon Plasma Properties.” Poster presentation at the PHENOEXP Workshop, August 2026. [\[link\]](#).
2. “Study of the Effect of Correlated Experimental Uncertainty on Bayesian Inference of Quark–Gluon Plasma Properties.” Oral presentation at the XXXIII Reunião de Trabalho sobre Interações Hadrônicas (RETINHA XXXIII), June 2026. [\[link\]](#).
3. “Risk Management and IT Governance in the Adoption of Quantum Computing: A COBIT 5-Based Conceptual Model.” Oral presentation at the 5th Congress on Risk Management and ERM, FEA–USP, December 2025. [\[link\]](#).
4. “Exploring the Potential of Quantum Computing in Materials Science.” Oral presentation (co-author), Autumn Meeting of the Brazilian Physical Society, May 2025. [\[link\]](#).

Industry & Professional Talks

1. “Unveiling Quantum Computing: Practical Applications and Future Possibilities.” Talk at Rio Innovation Week – AI and Meta World, 2024. [\[link\]](#).
2. “Quantum Computing. Hands On: Grover Search Algorithm.” Talk at Open Co, Tech Day, 2022. *[Internal material]*
3. “Shift Your Programming Paradigm to a Quantum Mindset.” Talk at The Developer’s Conference (TDC Business), 2022. [\[link\]](#).
4. “Fundamentals of Quantum Computing.” Talk at Open Co, Tech Day, 2021. *[Internal material]*
5. “Technical Software Documentation.” Seminar at ITAUTEC, 2014. *[Internal material]*

Early Education Highlights

School Environmental Research Project – Municipal Waste Management	1999
Colégio Galáxia – Goiânia, Goiás, Brazil	
- Conducted a school research project on municipal solid waste management, including an interview with Pedro Wilson Guimarães, then Federal Deputy for the State of Goiás. - Investigated waste-management challenges and technologies, including landfill biogas recovery and international approaches to waste incineration. - Wilson later served as Mayor of Goiânia (2001–2004), where his administration established municipal selective waste collection through legislation enacted in 2002.	
School Environmental Field Visit – Goiânia Municipal Landfill [link]	1998
Escola do Parque [link] – Goiânia, Goiás, Brazil	
- Participated in a field visit to the Goiânia Municipal Landfill, observing municipal solid waste disposal operations and environmental-control systems. - Visited the landfill-gas collection and flaring system, where biogas was captured and burned rather than recovered for energy production at the time. - Observed the leachate treatment system used to treat landfill effluent before the treated water was discharged back into the local river system.	

	“Aprender a Pensar” – High Abilities/Gifted Education Extension Program Pontifical Catholic University of Goiás (PUC Goiás) – Goiânia, Goiás, Brazil • Selected for participation in a university extension program for children identified with high abilities/giftedness through psychometric assessment of non-verbal abstract reasoning, including visual pattern recognition and inductive reasoning tasks. • Participated in enrichment activities designed to develop higher-order reasoning, problem-solving, creativity, and metacognitive skills.	1996
	Logosophical Education – First Grade Colégio Logosófico González Pecotche – Goiânia, Goiás, Brazil • Completed first grade at a school whose educational approach was grounded in Logosophy, a system of thought developed by Carlos Bernardo González Pecotche that emphasizes conscious self-knowledge and the development of individual capacities. • The educational experience placed particular emphasis on activities fostering self-awareness, reflection, creativity, and personal expression.	1995
Certificates	HSK 3 – Chinese Proficiency Exam Scheduled [link] Instituto Confúcio / São Paulo, SP	18 Oct 2026
Extracurricular Activities	1st Place – Brazil Interstate Taekwon-Do ITF Competition [link] • Two Gold medals in Sparring and Patterns (Tul). • Buzios. Rio de Janeiro, Brazil. 1st Place – Children’s Drawing Contest [Art Portifolio] Claro – Telecommunications Company, Brazil • First-place award in a Children’s Day drawing competition organized by Claro for employees’ children.	Aug 2023 Oct 2000